

Ting-Hsuan Chen

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📍 Los Angeles (U.S. Citizen) ✉ tchen783@usc.edu ☎ (323)791-8418 🔗 koi953215.github.io 📧 koi953215

Education

University of Southern California

Aug 2024 – present

PhD in Computer Science; MS in Computer Science | GPA: **3.92/4.0**

National Chung Hsing University

Sept 2019 – June 2023

Department of Applied Mathematics (DAM) - Computer Science Group

- Final Average **96.6%**, GPA **4.27/4.3**; ranked 1st in all eight semesters; Valedictorian
- Admitted as the representative of the College of Science to the Phi Tau Phi Honor Society

Experience

Scene Understanding/GenAI – Research Intern

Sunnyvale, United States

Bosch Center for Artificial Intelligence

May 2025 – present

- Proposed a 3D-aware 360° video diffusion framework that leverages an explicit point cloud cache to ensure geometric consistency and precise trajectory control for digital twin generation.
- Engineered a diffusion-based planning system with DINOv2 perception, enabling robust zero-shot generalization in challenging environments.

PhD student/Research Assistant

Los Angeles, United States

University of Southern California

Aug 2024 – present

Advised by Prof. Yue Wang. My research turns generative priors into physical intelligence, along three thrusts:

- **Data scaling**: using video diffusion and world models to generate robot data beyond teleoperation.
- **Physical consistency**: pushing generated content to obey real-world physics, not merely look plausible.
- **World-action coupling**: how world models turn prediction into action, via inverse dynamics, joint video-action denoising, or representation transfer.

MS student (Spring 2024)/Research Assistant

Hsinchu, Taiwan

National Yang Ming Chiao Tung University

Jan 2024 – June 2024

- Entered Prof. Yu-Lun Liu's laboratory and led a Google-sponsored industry-academia collaboration project.
- Collaborated with MediaTek to develop the first Zero-Shot Video Restoration framework utilizing diffusion-based image restoration models.

R&D Engineer

Taipei, Taiwan

Foxconn, Hon Hai Precision Industry

July 2023 – Dec 2023

- Developed the company's first patented ECG waveform recognition system within three months, integrating cutting-edge AI, computer vision, and signal processing technologies.
- Served as a technical mentor for interns and represented the company at medical conferences.

Publications (*: Equal contribution, †: Co-corresponding authors)

TimePre: Bridging Accuracy, Efficiency, and Stability in Probabilistic Time-Series Forecasting

[TMLR](#)

- Lingyu Jiang*, Lingyu Xu*, ..., [Ting-Hsuan Chen](#), ..., Michael Zielewski, Kazunori Yamada, Fangzhou Lin

Pantheon360: Taming Digital Twin Generation via 3D-Aware 360° Video Diffusion

[CVPR 2026](#)

- [Ting-Hsuan Chen](#)*, Ying-Huan Chen*, ..., Yuliang Guo†, Yu-Lun Liu†, Yue Wang†, Liu Ren

Dino-Diffusion Modular Designs Bridge the Cross-Domain Gap in Autonomous Parking

[under review](#)

- Zixuan Wu, Hengyuan Zhang, [Ting-Hsuan Chen](#), Yuliang Guo, David Paz, Xinyu Huang, Liu Ren

NaRCan: Natural Refined Canonical Image with Integration of Diffusion Prior for Video Editing

[NeurIPS 2024](#)

- [Ting-Hsuan Chen](#), Jiewen Chan, Hau-Shiang Shiu, Shih Han Yen, Changan Yeh, Yu-Lun Liu

DiffIR2VR-Zero: Zero-Shot Video Restoration with Diffusion-based Image Restoration Models

[arXiv 2024](#)

- Changan Yeh, Chin-Yang Lin, Zhixiang Wang, Chi-Wei Hsiao, [Ting-Hsuan Chen](#), Yu-Lun Liu